

Zomato Dataset Analysis Report

Alfido Tech Internship – Task 1

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1. Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Zomato dataset to identify trends, customer preferences, restaurant ratings, pricing patterns, and online delivery services.

2. Dataset Description

The dataset contains information related to restaurants listed on Zomato. It includes restaurant names, ratings, votes, location details, online delivery availability, table booking options, and approximate cost for two people.

3. Data Cleaning and Preprocessing

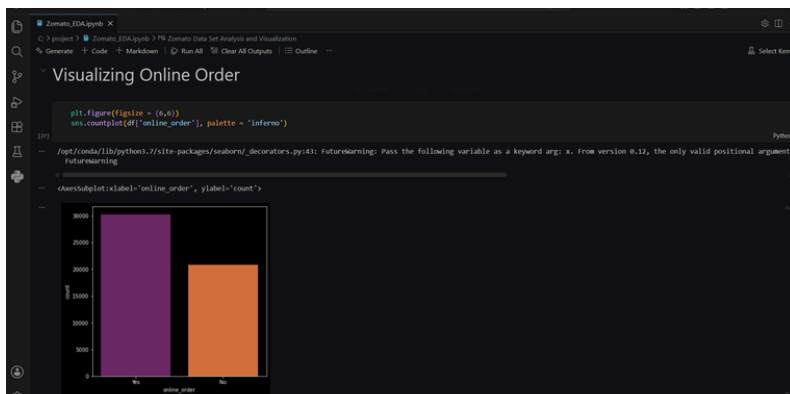
Several preprocessing steps were performed before analysis:

- Missing values were identified and handled.
- Duplicate records were checked and removed if necessary.
- Data types were verified and corrected.
- Irrelevant columns were removed for better analysis.
- Column names were standardized for readability.

4. Exploratory Data Analysis

Different visualization techniques were used to understand the dataset:

- Rating distribution analysis
- Online delivery analysis
- Cost distribution analysis
- Restaurant type analysis
- Votes and customer engagement trends



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No. of Votes, Location Wise

```
df4 = df[['location', 'votes']]
df4.drop_duplicates()
df5 = df4.groupby(['location'])['votes'].sum()
df5 = df5.to_frame()
df5 = df5.sort_values('votes', ascending=False)
df5.head()
```

```
[38]
```

location	votes
Koramangala 5th Block	2214083
Indiranagar	1165909
Koramangala 4th Block	685156
Church Street	590306
JP Nagar	586522

```
plt.figure(figsize = (15,8))
sns.barplot(df5.index , df5['votes'])
plt.xticks(rotation = 90)
```

```
[39]
```

/opt/conda/lib/python3.7/site-packages/seaborn/_decorators.py:43: FutureWarning: Pass the following variables as keyword args: x, y. From version 0.12, the only valid positional argument will be x. See https://matplotlib.org/3.1.1/api/_as_gen/plt.bar.html for more information.

```
(array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
        17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
        34, 35, 36, 37, 38, 39, 40, 41]))
```

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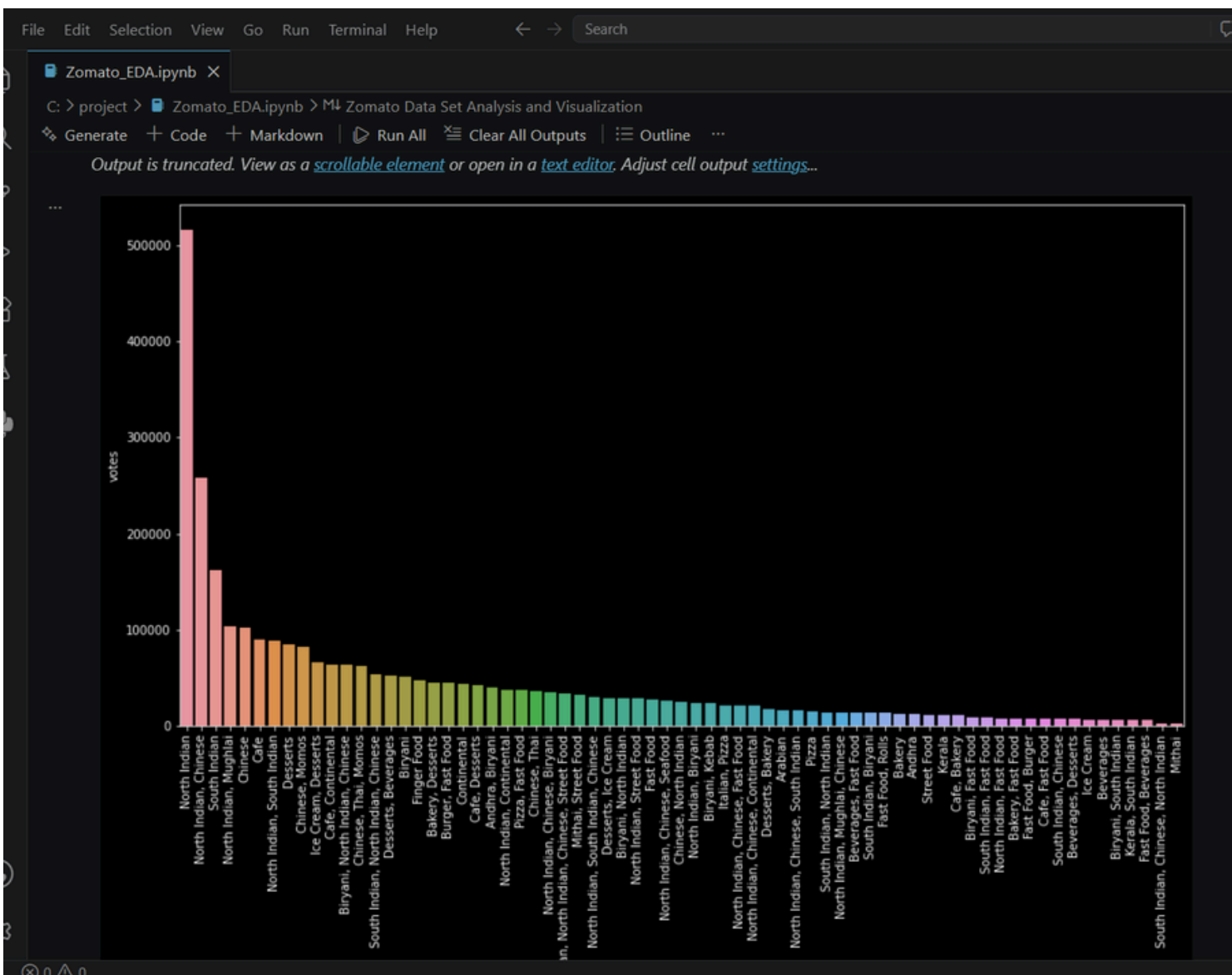
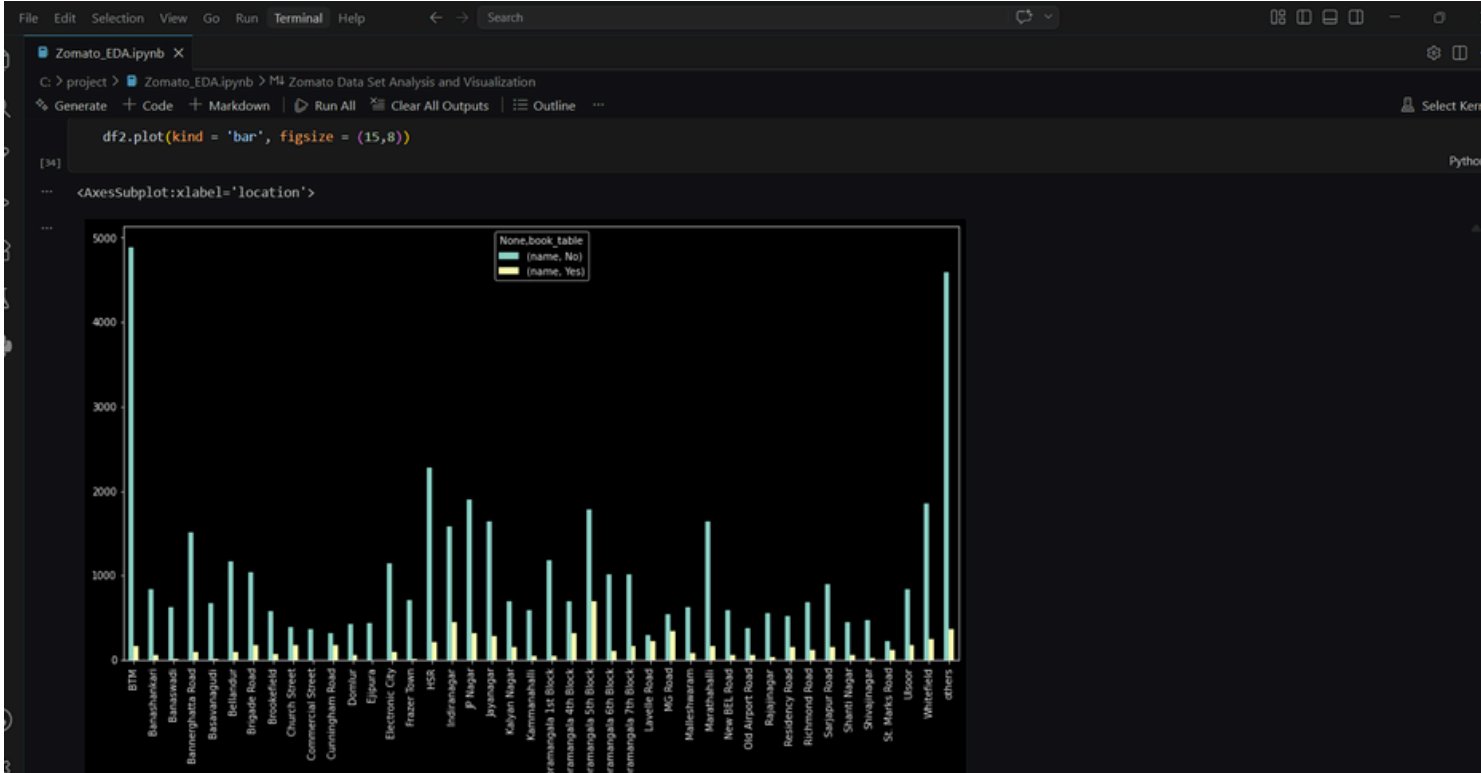
```
df.head()
```

```
[40]
```

	name	online_order	book_table	rate	votes	location	rest_type	cuisines	Cost2plates	Type
0	Jalsa	Yes	Yes	4.1	775	Banashankari	Casual Dining	North Indian, Mughlai, Chinese	800.0	Buffet
1	Spice Elephant	Yes	No	4.1	787	Banashankari	Casual Dining	others	800.0	Buffet
2	San Churro Cafe	Yes	No	3.8	918	Banashankari	others	others	800.0	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	Banashankari	Quick Bites	South Indian, North Indian	300.0	Buffet
4	Grand Village	No	No	3.8	166	Basavanagudi	Casual Dining	others	600.0	Buffet

Visualizing Top Cuisines

```
df6 = df[['cuisines', 'votes']]
df6.drop_duplicates()
df7 = df6.groupby(['cuisines'])['votes'].sum()
df7 = df7.to_frame()
df7 = df7.sort_values('votes', ascending=False)
```



5. Key Insights

- Most restaurants have ratings between 3.5 and 4.0.
- Online delivery services are widely used among restaurants.
- Mid-range restaurants are more common compared to premium restaurants.
- Restaurants with higher ratings generally receive more customer votes.
- Casual dining and cafe categories dominate the dataset.

6. Conclusion

The analysis provided meaningful insights into restaurant trends and customer preferences using the Zomato dataset. EDA techniques and visualizations helped identify important business patterns related to ratings, pricing, and online delivery.